

Vacancy-Confined Single-Atom Catalysts on MXenes for Enhanced CO₂ Reduction

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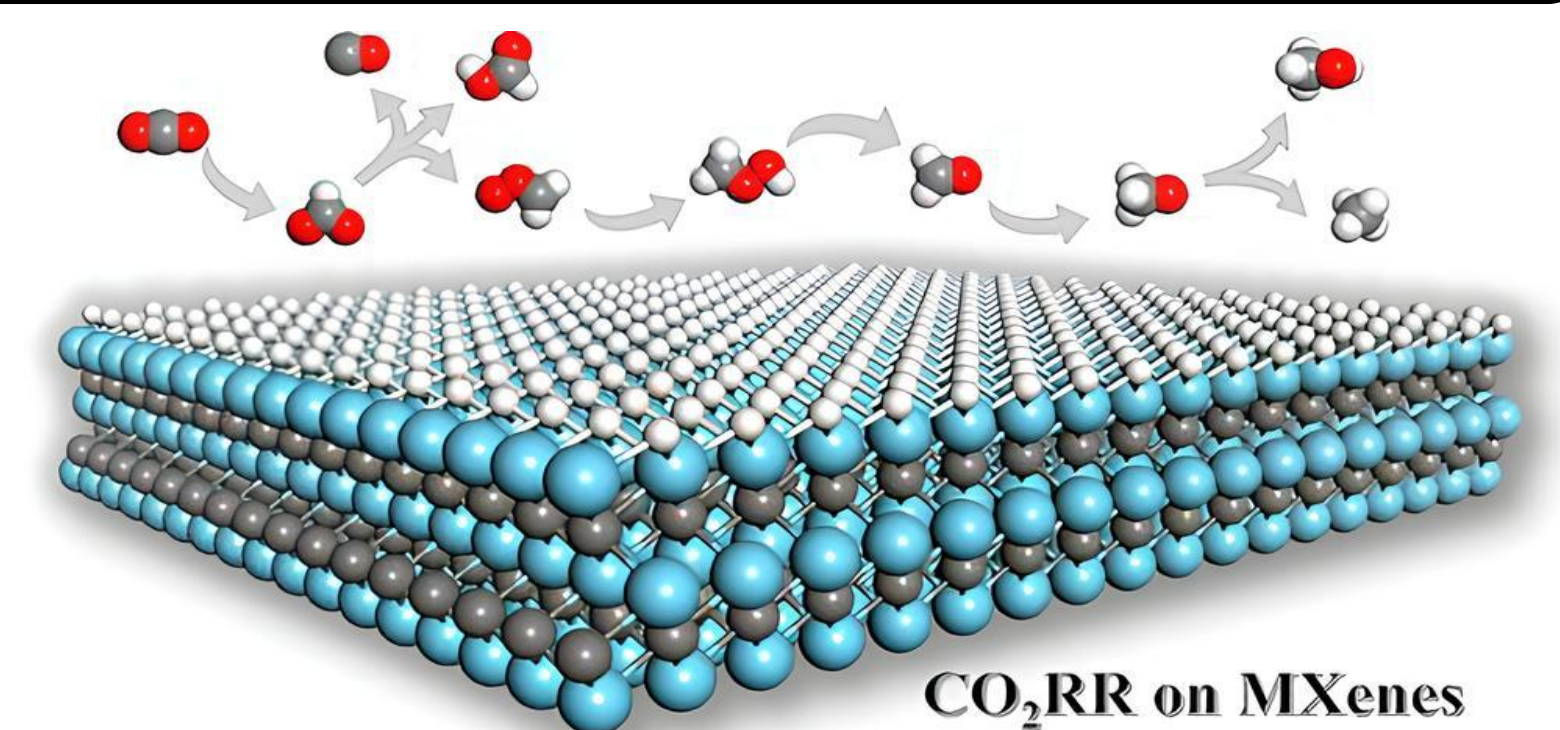
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Abstract

Vacancy-confined single-atom catalysts on MXenes have emerged as a promising strategy for boosting the efficiency of electrochemical CO₂ reduction. The strong interaction between single metal atoms and engineered vacancy sites enhances atom stability, optimizes electronic structure, and provides abundant active centers for selective CO₂ conversion. This work highlights advances in vacancy engineering of MXenes to anchor isolated atoms, elucidates structure-activity relationships, and discusses the potential of these materials in achieving high catalytic activity, stability, and product selectivity. Such approaches offer a pathway toward sustainable and scalable CO₂ utilization technologies.



Introduction

Achieving efficient CO₂RR, requires catalysts with high conductivity, tunable surface chemistry, and structural versatility. MXenes, a family of 2D transition metal carbides and nitrides derived from MAX phases, have recently attracted attention owing to their rich compositional diversity and unique physicochemical properties [1]. The broad range of available transition metals, A-group elements, and X-site atoms used to design tailored MXene-based catalytic platforms is illustrated in Figure 1.

H																			He
Li	Be																		
Na	Mg																		
K	Ca																		
Rb	Sr																		
Cs	Ba																		
Fr	Ra																		

Figure 1. Periodic table with MAX/MXene, M, A and X site elements².

Preparation Method

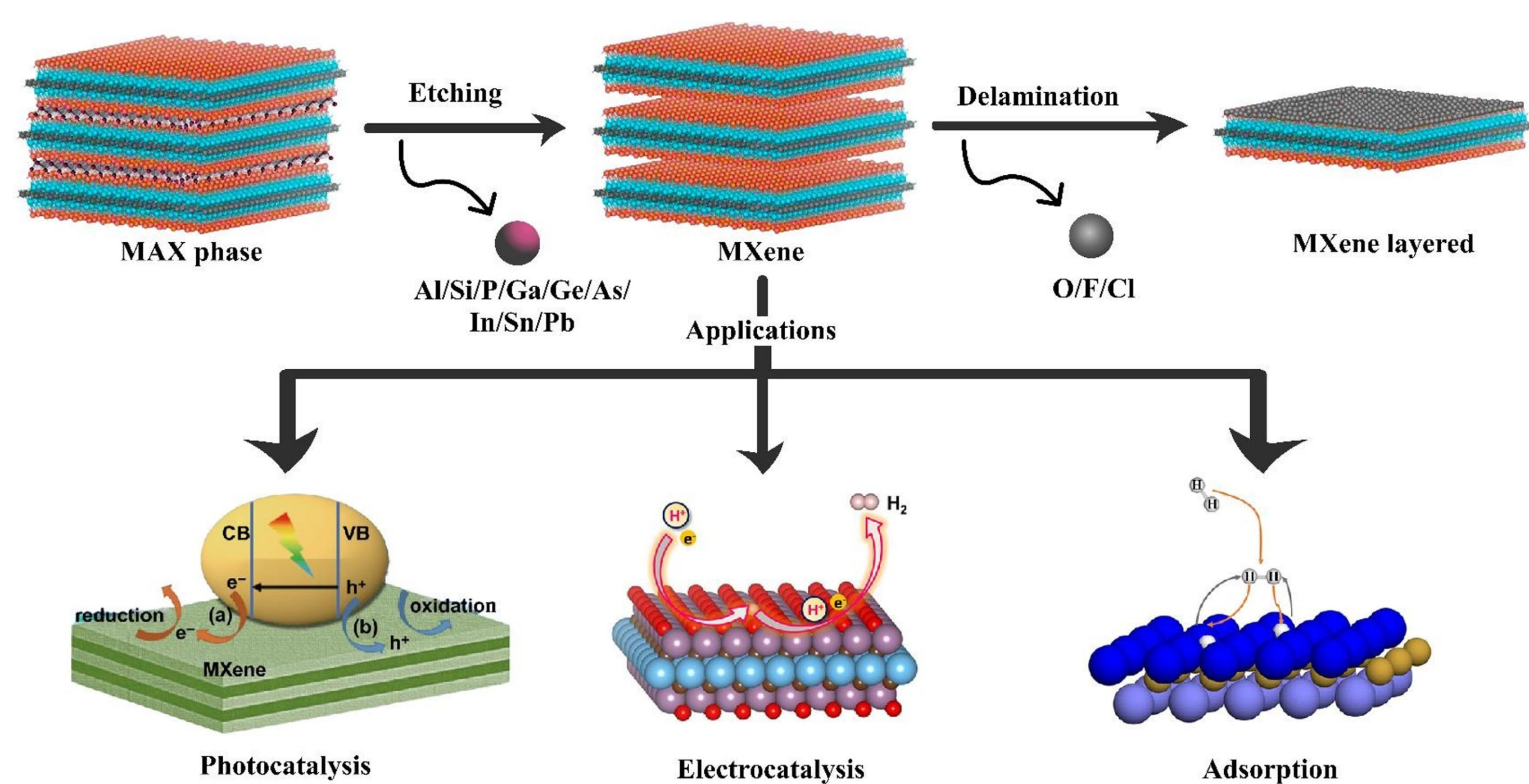


Figure 2. MAX to MXene etching and some common uses³.

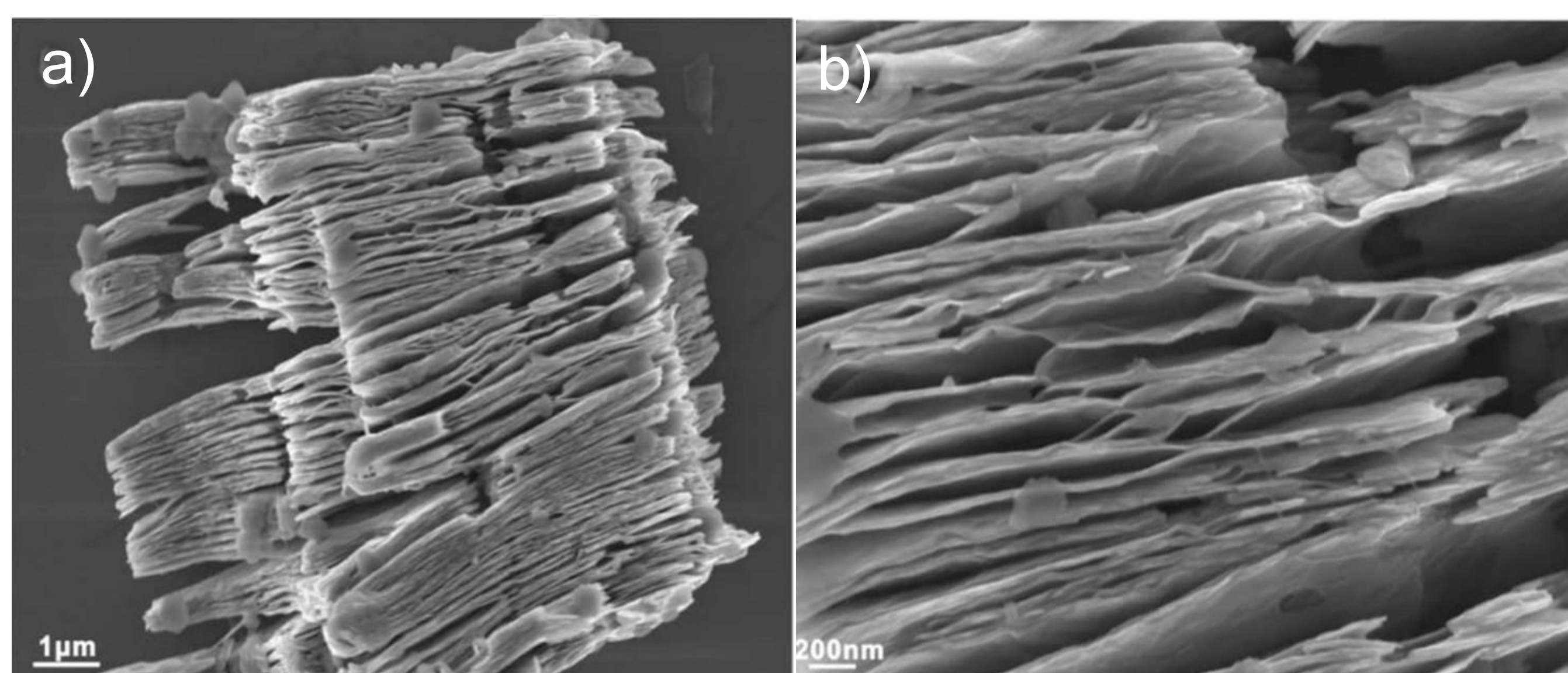


Figure 3. The SEM image of Ti₃C₂T_x (a) (b) under different magnification.

Results

Adsorption configurations of various intermediates were determined using *density functional theory* (DFT) calculations within the *VASP* software package.

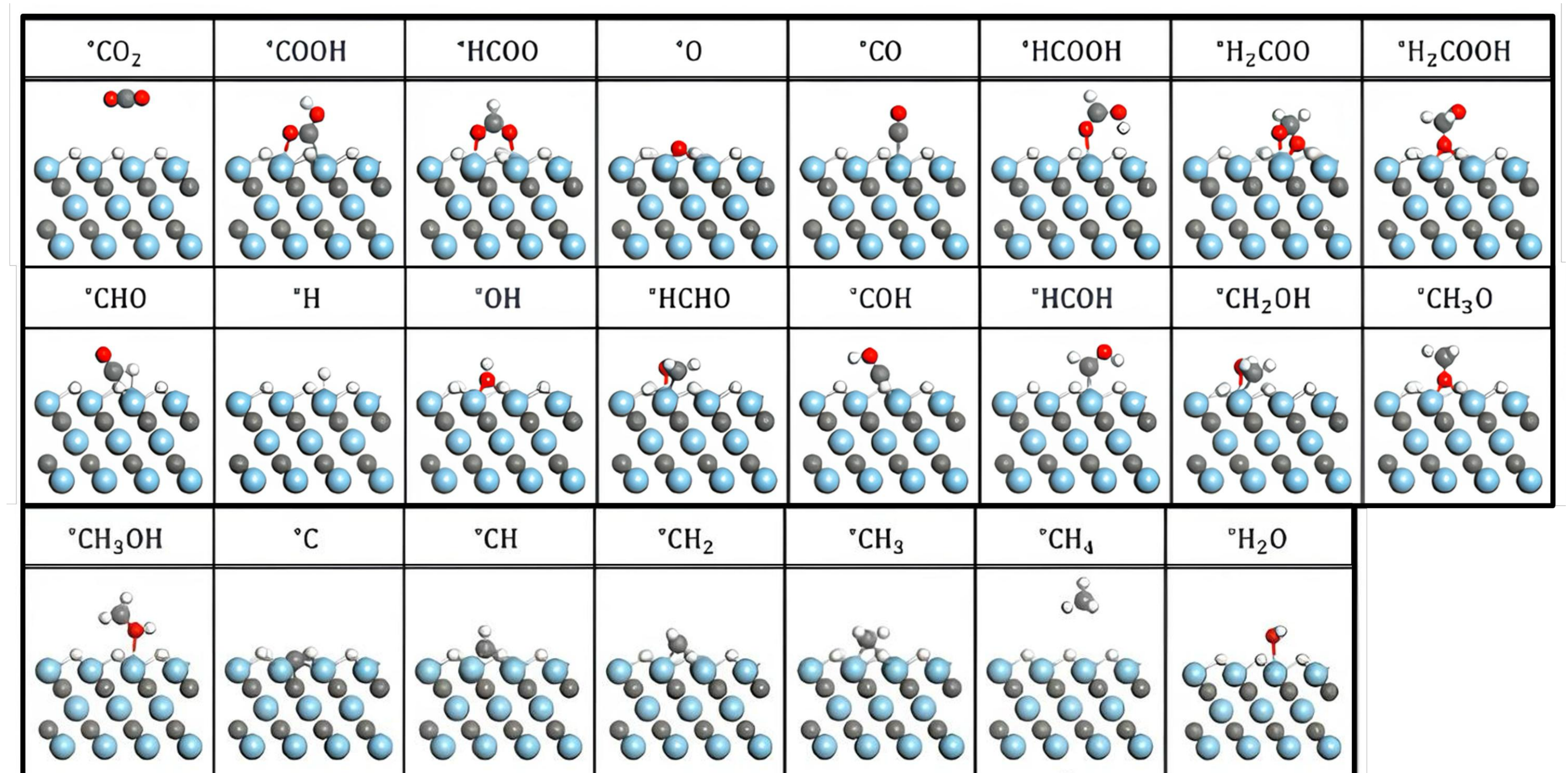


Figure 4. Adsorption configurations of various intermediates and products of CO₂ reduction on the MXene surface. .

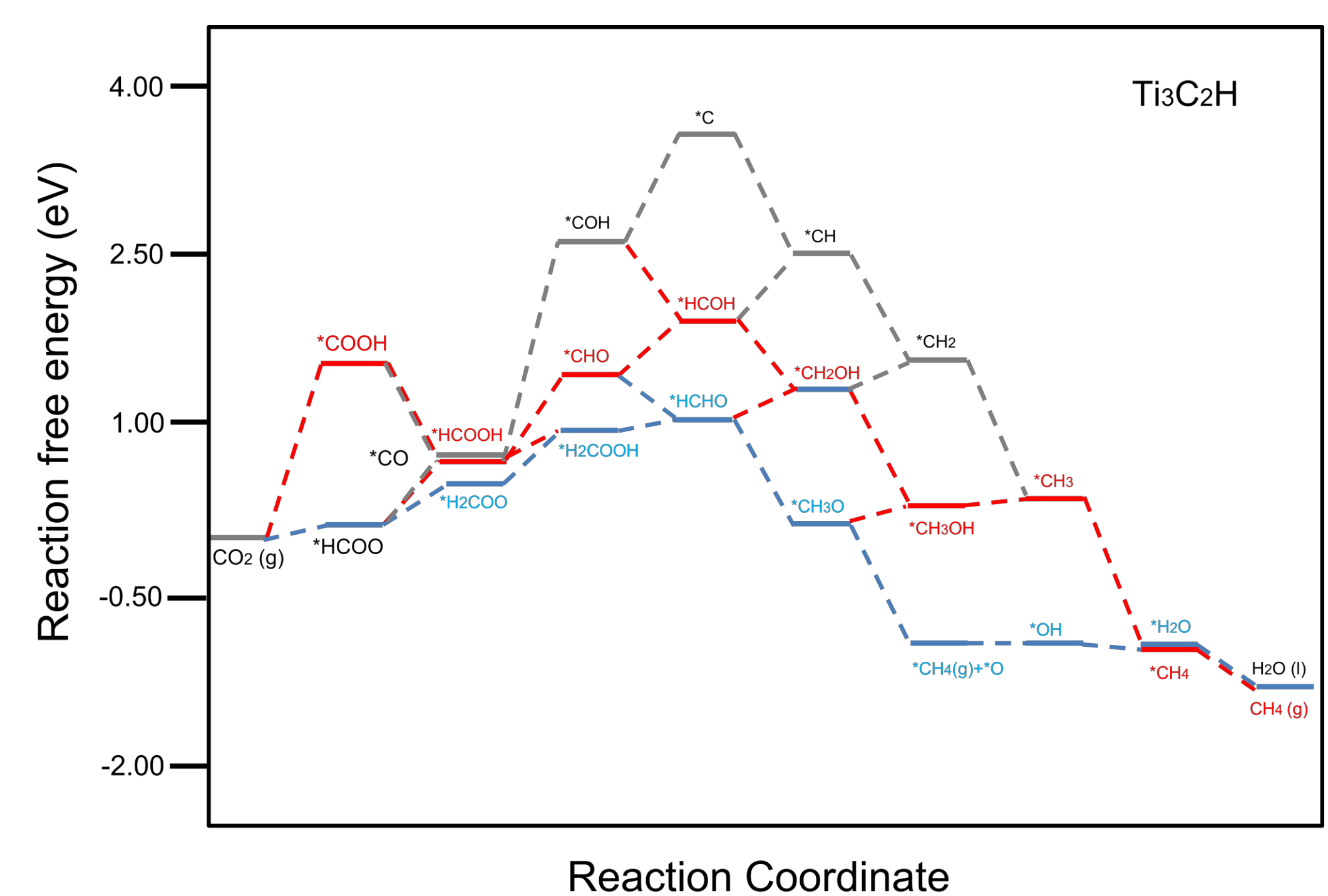


Figure 5. Reaction free energy diagram for the electrochemical reduction of CO₂ to different products, including CO (blue pathway), CH₄ (red pathway), and H₂O (gray pathway). The diagram illustrates the energy barriers and stability of key intermediates, providing insights into the reaction mechanism.

Concluding Remarks

Vacancy engineering in MXenes provides a powerful strategy to stabilize single atoms and optimize their catalytic environment for CO₂ reduction. This approach opens new opportunities for designing efficient, selective, and durable catalysts toward sustainable carbon utilization.

References

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